

Chapter 15 — Production

Documents, figures and diagrams

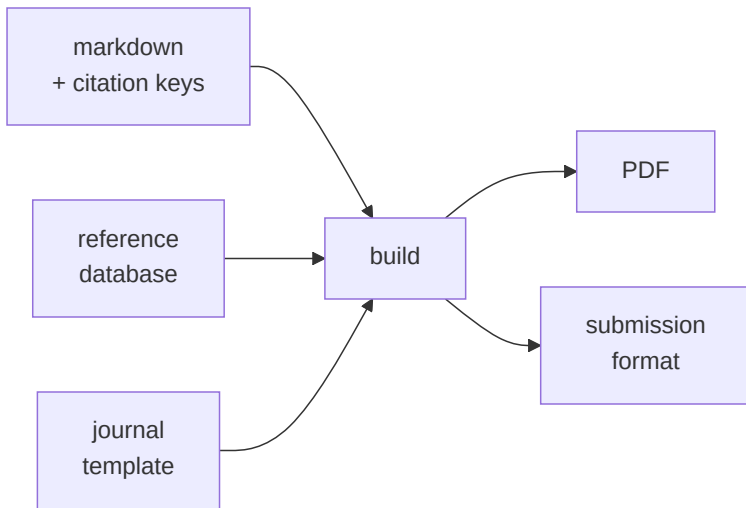
Everything in this chapter is a text file under version control that renders into something a journal will accept.

Why a text pipeline beats a word processor

- It **diffs**. You can see what changed between drafts, and so can your supervisor.
- It is **reproducible**. The same source and the same template give the same document, on any machine.
- **Citations are data, not typing**. A bibliography is generated from identifiers, which is the Chapter 14 habit made mechanical.
- **Templates are swappable**. The same manuscript retargets to a different journal without reformatting by hand.

The cost is real: a build step, and a day of setup. Say that plainly rather than pretending the trade is free.

The pipeline in one line



- Cross-references, figure numbering and citation styling are handled by the build, not by you.
- The same source produces the preprint, the submission and the slides.

Figures that can be regenerated

- A figure produced by a committed script can be audited, corrected and rebuilt. A figure pasted from a screenshot cannot.
- Where a figure has an equation behind it, the script is the derivation, and it is reviewable.
- Where input values are illustrative rather than measured, **say so on the figure itself**, not in the caption of a draft that will be edited.

This deck follows the same rule. Its computed figures come from a script in the repository, and the illustrative inputs are labelled in the image.

Diagrams as text

- Flowcharts and system maps written as text render into diagrams, diff like code, and survive export.
- The alternative — a drawing tool — produces a binary nobody else can edit and nobody can review.
- For anything with boxes and arrows, text wins. For anything with data, plot it instead.

And for anything else: generated imagery has no provenance and cannot be regenerated deterministically. In a document that carries claims, that is the wrong trade.

ASCII output has a real use

- Terminal output, code comments, commit messages, plain-text README files.
- A diagram inside a source file stays with the code it describes and needs no build step to read.
- Small, ugly and durable beats beautiful and detached.

Explaining your own code to other people

- A supervisor, a collaborator, or you in eighteen months.
- Ask for the explanation, then check it against what the code does, not against what you meant it to do. The gap between those two is the useful output.
- Where the explanation is wrong, that is usually the code being unclear rather than the reader being wrong.

Where the explanation is right, it goes in the project instruction file from Chapter 10, and stops being re-derived every session.

Exercise

1. Take one section of something you are writing. Put it through the pipeline end to end, with one real citation.
2. Retarget it to a second template and confirm nothing broke.
3. Regenerate one figure from a script and commit the script beside it.

If step 2 breaks, you have found the reason to do this now rather than the week before a deadline.